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NumPy Array Shape

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Shape of an Array

The shape of an array is the number of elements in each dimension.

Get the Shape of an Array

NumPy arrays have an attribute called **shape** that returns a tuple with each index having the number of corresponding elements.

Example

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Print the shape of a 2-D array:

```
import numpy as np

arr = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])

print(arr.shape)
```

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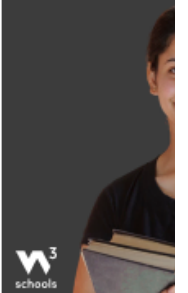
The example above returns **(2, 4)**, which means that the array has 2 dimensions, where the first dimension has 2 elements and the second has 4.

Example

Create an array with 5 dimensions using **ndmin** using a vector with values 1,2,3,4 and verify that last dimension has value 4:

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```
import numpy as np

arr = np.array([1, 2, 3, 4], ndmin=5)

print(arr)
print('shape of array :', arr.shape)
```

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What does the shape tuple represent?

Integers at every index tells about the number of elements the corresponding dimension has.

In the example above at index-4 we have value 4, so we can say that 5th (4 + 1 th) dimension has 4 elements.

Exercise ?

Consider the following code:

```
import numpy as np
arr = np.array([[1, 2, 3], [4, 5, 6]])
print(arr.shape)
```

What will be the printed result?

- ☐ (3, 2)
- ☐ (2, 3)
- ☐ (2, 2)
- ☐ (3, 3)

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